LOCAL FINITE PRESENTATION OF A COMPOSITE MORPHISM OVER A LOCALLY FINITE TYPE INTERMEDIATE SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has no numbered parts; the proof is the standard chain-of-presentations argument and it is complete.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting a chain of morphisms of algebraic spaces to a chain of schemes. used at proof lines 5836-5839. Produces W, V, U with surjective etale morphisms onto Z, W x_Z Y and V x_Y X.
- The definitions of locally of finite presentation and locally of finite type for algebraic spaces. used at proof lines 5839-5840. Translate the hypotheses into statements about U over W and V over W.
- The scheme case of permanence of local finite presentation. Morphisms, Lemma morphisms-lemma-finite-presentation-permanence, used at proof lines 5840-5842. Gives that U -> V is locally of finite presentation.
- Local finite presentation is read off from a presentation. used at proof lines 5842-5843. Concludes for f.

Source reference. Original lemma 02FV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3206–3206. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ and $Y \to Z$ be morphisms of algebraic spaces over S. If X is locally of finite presentation over Z, and Y is locally of finite type over Z, then f is locally of finite presentation.*

Proof. Choose a scheme W and a surjective étale morphism $W \to Z$. Then choose a scheme V and a surjective étale morphism $V \to W \times_Z Y$. Finally choose a scheme U and a surjective étale morphism $U \to V \times_Y X$. By definition U is locally of finite presentation over W and V is locally of finite type over W. By Morphisms, Lemma 02FV the morphism $U \to V$ is locally of finite presentation. Hence f is locally of finite presentation. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.